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Medicine model

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Medicine concentrations

Simulate how the medicine in a pill someone takes enters into the bloodstream, does its work and is filtered out again.

Medicine model

In the following equations the quantities of medicine in the GI tract at time t are given by $M_{GI}(t)$ and $M_{blood}(t)$ respectively. V_{GI} is the volume of the contents in the GI tract and V_{blood} is the volume of blood of the person taking the medicine.

$$\begin{aligned}\frac{dM_{GI}}{dt} &= -d_1 \left(\frac{M_{GI}(t)}{V_{GI}} - \frac{M_{blood}(t)}{V_{blood}} \right) \\ \frac{dM_{blood}}{dt} &= d_1 \left(\frac{M_{GI}(t)}{V_{GI}} - \frac{M_{blood}(t)}{V_{blood}} \right) - d_2 \frac{M_{blood}(t)}{V_{blood}}\end{aligned}$$

Tasks for the project in Module 4

Finish the modelling for the full basic model. First a repeat of the optional tasks in module 3; if you have not done them there, do them here.

1. Implement Euler's method for the system for $M_{GI}(t)$ and $M_{blood}(t)$. If you have decided on different stepsizes for the calculations in items 4 and 5, for the system you

need to calculate with the minimum of the two stepsizes. If in doubt check the errors again.

2. Check again the values of d_1 and d_2 . Are the results plausible?

New task for module 4:

3. Make a graph of solutions $M_{GI}(t)$ for different initial conditions. Use not only different values for $M_{GI}(0)$, but also for $M_{blood}(0)$. Do you understand the relation between the initial conditions and $M_{GI}(t)$? Do the same for $M_{blood}(t)$.
4. Graph the same solutions for different sets of initial conditions in the phase plane, the $M_{GI} M_{blood}$ -plane. Is the type of the equilibrium point as expected?
5. Simulate taking a pill every x hours. You do that by checking whether it is medicine time with an if-statement `if np remainder(t,x)< small number (smaller than the step size):` and then you add the amount of a pill to M_{GI} : $M_{GI} = M_{GI} + \text{Pill}$. Simulate for longer times and generate graphs of $M_{GI}(t)$ and $M_{blood}(t)$ versus time.
6. For fun, you can also plot the solutions of task 5 in the phase plane, by plotting M_{blood} versus M_{GI} . Do you understand the result? Is the set of differential equations solved in task 5 autonomous?

Extending your model

After you have written and submitted your first report, in module 5 you have to go beyond the basic model. You should start thinking about what you would like to do then. What specific problem about the medicines do you want to solve? That can be a pressing real-life problem, or you could investigate something you made up. What if ...?

An important real-life question is how often should you take the medicine in what quantity to get a stable enough concentration in the blood. And what happens when you are not very strict in taking the pills, and you forget one sometimes. Should you take a double dosis the next time?

What is the difference between taking the pill after a meal, or on an empty stomach? What about slow release pills?

You could also simulate for a breastfeeding mother. The milk that the baby drinks is mostly made a la minute from nutrients in the blood of the mother, and medicine diffuses from the blood into the milk as well. So you could alter the second differential equation and add a third differential equation for the amount of medicine in the milk that the baby drinks. Try to find a pill taking regime that is good enough for the mother, and lets the baby get as little as possible.

You may be dissatisfied with this model for your the medicine concentrations, and for yet another extension of the problem, you may alter the terms, such that they model your medicine diffusion better. Then your modelling problem could be What is a mathematical model to predict ...?

This extension of the model becomes relevant only in module 5. Finish your work on the basic model first and write and submit the first report before June 27th.

The next page is for the questions you might have about the project.

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